Griffiths 2021 Breast Cancer — analysis summary

RUN 5 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Two platform leaves (10x 110,568 + iCell8 804 malignant) share zero samples/patients and never mix under Harmony (platform iLISI ~1.0 across the sweep) → trajectory scoped to the 10x leaf. On the 10x leaf, 71 ER+ tumours intermix on one continuous manifold (sample iLISI 1.21→12.63 of 71; independent Harmony-PCA kNN = 1 component; 110,050 cells retained). Cycling is orthogonal to the embryonic axis → whole manifold kept. GO.

Gate 2 — module geography → —

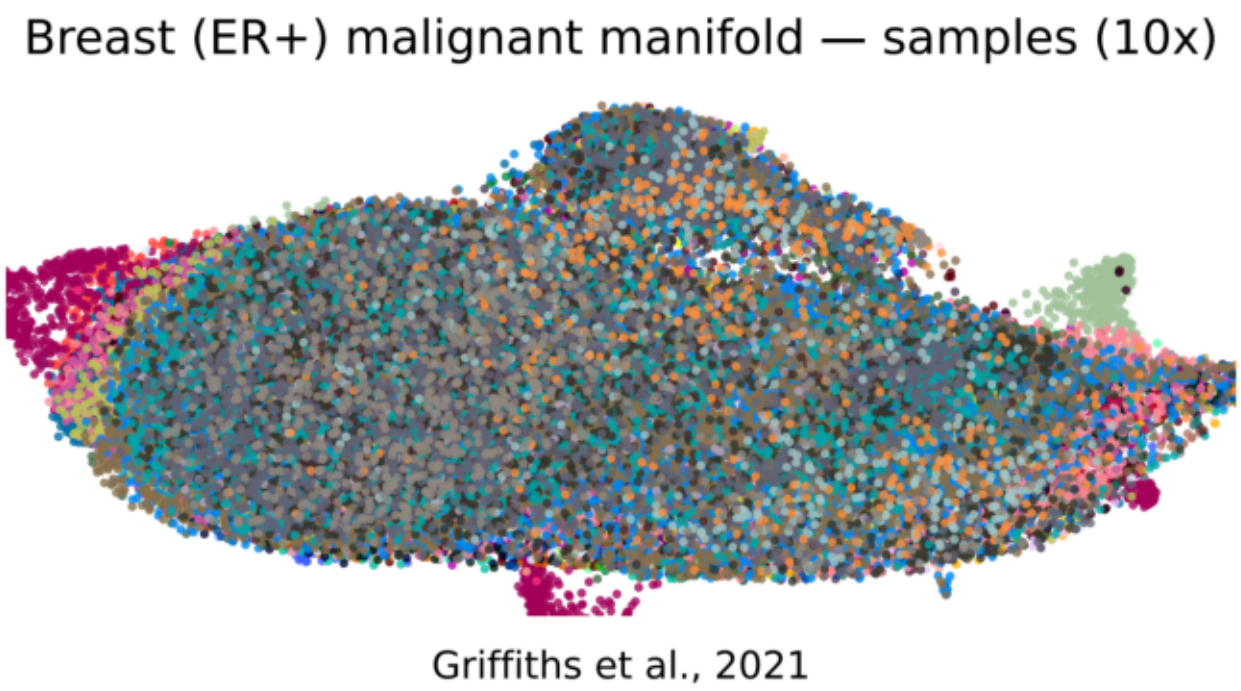
Not evaluated — this attempt stopped after Gate 1 (only 01_preprocessing produced; no trajectory / signature steps committed, no processed.h5ad).

Gate 3 — hub cross-repression → —

Not evaluated — downstream of Gate 2, which was not reached.

STEP FIGURES

Manifold — samples (Gate 1)



Dataset.

Breast Cancer (ER+, FELINE neoadjuvant trial; Griffiths et al., 2021), 10x + iCell8 — 10x leaf scoped: 110,050 malignant cells across 71 samples (≥50 cells).

Verdict.

INCOMPLETE ATTEMPT — reached Gate 1 GO (71 samples form one continuous 10x manifold; cycling orthogonal) but stopped before the trajectory / signature steps; only the samples deliverable was produced. Gate 1 GO / Gates 2 & 3 not evaluated. Superseded by later Griffiths runs.

Parameter choices

Platform scope	10x leaf (iCell8 804 cells never co-embed; platform iLISI ~1.0)	**Sample threshold**	≥50 cells/sample (smooth dist, no clean cliff); retains >99.8%
Integration	Harmony on sample (iLISI-sweep); n_pcs=30 (scree)	**Island removal**	radius graph 6× median-NN, largest component → 110,050 cells
Gate 1	GO — sample iLISI 1.21→12.63 of 71; 1 connected component; cycling orthogonal	**Status**	INCOMPLETE — stopped after Gate 1 (preprocessing only)